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# Difficulty-Aware Language Routing and Cross-Lingual Self-Consistency for Korean Mathematical Reasoning

Korea University COSE461 Final Project

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| <b>Lee Yunje</b>           | <b>Kim Sangjun</b>         | <b>Won Junseo</b>          |
| Department of Data Science | Department of Data Science | Department of Data Science |
| Team 20                    | Team 20                    | Team 20                    |
| 2022320317                 | 2022320306                 | 2022320302                 |

## Abstract

Small language models reason markedly better in English than in Korean on mathematical word problems. We trace part of this gap to the distillation pipeline itself: even a strong teacher (Gemini) produces lower-quality chain-of-thought (CoT) in Korean (85.7% correct) than in English (97.5%), so naively distilling Korean CoT transfers a noisier signal. We address this with a two-stage cross-lingual framework. At the **data level**, our main contribution **Difficulty-Aware Language Routing (DALR)** decides, per problem, whether to learn from a Korean or an English teacher CoT based on teacher correctness: English CoT is used only as a “bridge” on problems the Korean teacher fails. A controlled ablation (DALR vs. random English placement) shows DALR’s gain stems from *where* English reasoning is routed, not from added data. At the **inference level**, we apply **Cross-Lingual Self-Consistency (XLSC)**, sampling each problem in both Korean and English from the same DALR-trained model and aggregating by majority vote. On HRM8K (Korean) with Qwen2.5-3B, DALR achieves the best single-model Korean accuracy (61.79%), and DALR + XLSC reaches 69.90% (+16.00 pts over base); the XLSC inference-time gain reproduces on Llama-3.2-3B (+8.19) and EXAONE-3.5-2.4B (+8.79).

## 1 Introduction

Large language models demonstrate strong multi-step mathematical reasoning, but small models (e.g. 3B parameters) lag far behind, and the gap is worse outside English [1]. For Qwen2.5-3B-Instruct, the model solves **71.65%** of English problems (GSM8K [2]) but only **53.90%** of Korean problems (HRM8K [3]). Practical deployments often cannot afford large models yet must serve non-English users, so this gap is consequential.

A standard remedy is knowledge distillation: a strong teacher generates chain-of-thought (CoT) solutions [4], and a small student is fine-tuned on them. However, in the cross-lingual setting, the teacher itself is weaker in Korean: on our training set, Gemini produces correct CoT for 97.5% of English problems but only 85.7% of Korean problems. Distilling Korean CoT therefore transfers a noisier signal than English, leaving substantial room for improvement in Korean. Existing cross-lingual remedies act mostly at inference time, such as English-pivot translation or cross-lingual prompting [5, 6], and leave the student’s underlying weights monolingual; we instead intervene at the training-data construction stage so that the cross-lingual signal is internalized.

This motivates our central question: *can the teacher’s—and a model’s—stronger English reasoning be used to improve a small student’s Korean mathematical reasoning?* Our working assumption is that mathematical *logic* is largely language-agnostic, so connecting English reasoning patterns to Korean problems should help.

We pursue this through a two-stage cross-lingual framework. At the **data level**, our main contribution **DALR** routes each problem to a Korean or English teacher CoT based on teacher correctness. At the **inference level**, **XLSC** samples from the same DALR-trained model in both languages and aggregates by majority vote.

### Contributions.

- We propose **DALR**, a difficulty-aware data-construction method for cross-lingual CoT distillation, and show via a controlled ablation that its gains come from *routing*, not added data.
- We propose **XLSC**, an inference-time aggregation method that exploits the bilingual reasoning capability instilled by DALR.
- Empirically, DALR raises HRM8K accuracy from **53.90%** to **61.79%** on Qwen2.5-3B, and DALR + XLSC reaches **69.90%** (+16.00 pts over base); the XLSC inference-time gain reproduces on Llama-3.2-3B (+8.19) and EXAONE-3.5-2.4B (+8.79).

## 2 Related Work

**Chain-of-thought and its distillation.** Chain-of-thought (CoT) prompting elicits intermediate reasoning steps in large language models and substantially improves multi-step mathematical accuracy [4]. A subsequent line of work *distills* this ability into smaller students by fine-tuning them on teacher-generated CoT [7, 8, 9]. Our pipeline is an instance of CoT distillation, but unlike prior work that fixes a single teacher language for the whole training set, DALR chooses the CoT language *per problem* based on teacher correctness.

**Cross-lingual reasoning.** Reasoning transfers unevenly across languages: even strong models solve math problems markedly better in English than in lower-resource languages [1]. Existing remedies typically act at *inference* time, by translating non-English questions to English before reasoning (English-pivot) or by cross-lingual prompting [5, 6, 10], or at the *instruction-tuning* level with multilingual CoT data [11]. DALR instead acts at the *training-data construction* level: cross-lingual signal is selected per problem and internalized in the student weights, rather than appended at inference.

**Self-consistency.** Self-consistency samples multiple chains from one model and selects the majority answer [12], trading compute for accuracy. XLSC extends this idea *cross-lingually*: instead of  $N$  chains in a single language, it samples in both Korean and English from the same DALR-trained model and votes over the union, exploiting errors that decorrelate across languages [10].

**Data selection for distillation.** A complementary line filters teacher CoT by correctness, keeping only chains whose final answer matches the gold label [13]. DALR builds on this rejection-sampling view but treats teacher *language* as an additional axis of selection: when the Korean teacher fails, the English CoT is substituted as a “bridge,” so teacher failure itself serves as a per-problem difficulty signal.

**Parameter-efficient fine-tuning.** Across all setups we adapt the student via Low-Rank Adaptation (LoRA) [14], which injects trainable low-rank matrices into the frozen base model’s linear projections. This keeps per-setup training cost small enough to compare six configurations (A–E, E\_random) under matched hyperparameters, and isolates the effect of *which* CoT data is distilled—the focus of this paper—from confounds tied to full-finetuning capacity.

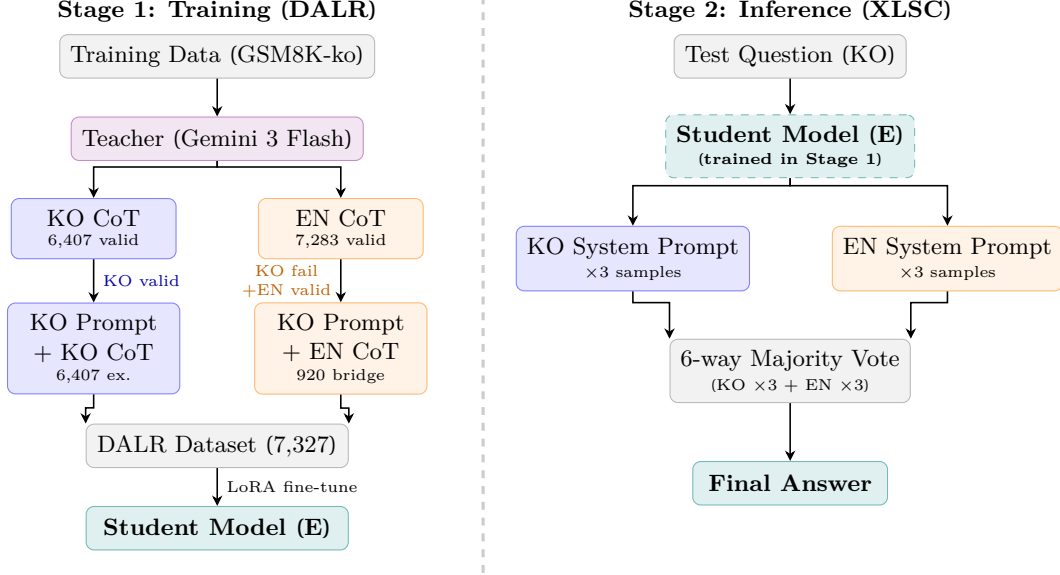


Figure 1: Two-stage cross-lingual framework. **Stage 1 (DALR)**: each training problem is routed to a Korean teacher CoT if valid, or an English “bridge” CoT otherwise; the 920 bridge examples target problems the Korean teacher fails. **Stage 2 (XLSC)**: at inference, the same student generates three Korean and three English chains per problem and aggregates by majority vote over the resulting six answers.

### 3 Approach

#### 3.1 Overview

We train a single small student with DALR-constructed data and aggregate its predictions at inference with XLSC. The two stages inject cross-lingual signal at different points: DALR at training time, deciding the CoT language per problem based on teacher correctness, and XLSC at inference time, sampling chains in both Korean and English from the same model and majority-voting. Figure 1 illustrates the framework end-to-end. The data routing in Stage 1 (Section 3.2) is our main methodological contribution; the inference-time aggregation in Stage 2 (Section 3.3) is a natural complement that exploits the bilingual capability instilled by Stage 1.

#### 3.2 Difficulty-Aware Language Routing (DALR)

The cross-lingual setting introduces a structural mismatch in CoT distillation: the teacher is unevenly competent across languages. On our Korean training set, Gemini produces a correct chain for only 85.7% of problems in Korean versus 97.5% in English, a 12-point gap. Distilling Korean CoT as-is therefore propagates a noisier signal to the student on the problems where the teacher itself struggles, presumably also where the student struggles—consistent with the per-difficulty analysis in Section 5. DALR turns this gap into a *free per-problem difficulty signal*: when the teacher fails in Korean, the problem is hard for Korean reasoning, so we substitute the English CoT as a higher-quality target rather than discarding the problem or fine-tuning on an unreliable Korean chain.

**Routing rule.** For each Korean training problem  $i$  we have the original Korean question  $q_i$ , a Korean teacher CoT  $c_i^{\text{KO}}$ , an English teacher CoT  $c_i^{\text{EN}}$ , and binary correctness flags  $v_i^{\text{KO}}, v_i^{\text{EN}} \in \{0, 1\}$  from exact-match on the final numeric answer. DALR emits exactly one

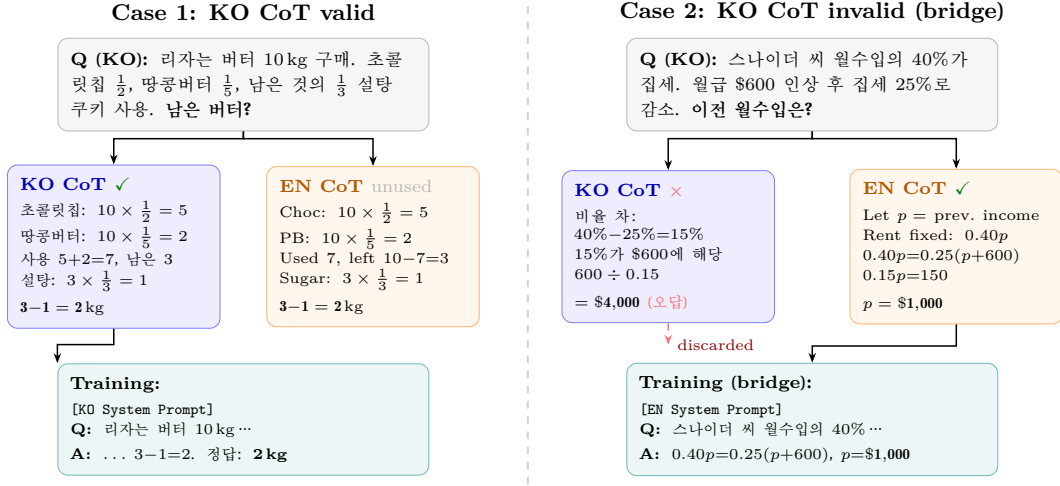


Figure 2: Concrete illustration of DALR routing (Eq. 1) on two GSM8K training problems. **Case 1** (Liza’s butter): the Korean teacher CoT correctly decomposes three fraction operations and reaches 2kg, becoming the direct training target. **Case 2** (Mrs. Snyder’s income): the Korean CoT conflates the 15-point percentage drop with the \$600 salary increase (\$4,000  $\neq$  \$1,000); the English CoT sets up the correct equation  $0.40p = 0.25(p+600)$  and is substituted as a bridge. The Korean question is unchanged in both cases.

training example  $t_i$  per problem:

$$t_i = \begin{cases} (q_i, c_i^{\text{KO}}, s^{\text{KO}}) & \text{if } v_i^{\text{KO}} = 1, \\ (q_i, c_i^{\text{EN}}, s^{\text{EN}}) & \text{if } v_i^{\text{KO}} = 0 \text{ and } v_i^{\text{EN}} = 1, \\ (\text{discarded}) & \text{otherwise,} \end{cases} \quad (1)$$

where  $s^{\text{KO}}$  and  $s^{\text{EN}}$  are the Korean and English system prompts respectively. The user question  $q_i$  is always Korean; only the CoT language and the system prompt change. The student is fine-tuned with the system prompt matching the target CoT language, so it learns to switch reasoning languages conditional on prompt rather than on question language—a property we exploit later for XLSC (Section 3.3). Figure 2 illustrates both routing cases on concrete GSM8K problems.

**Dataset.** Applied to GSM8K-ko’s 7,473 training problems, Eq. 1 yields 6,407 Korean-CoT examples ( $v^{\text{KO}}=1$ ), 920 English-bridge examples ( $v^{\text{KO}}=0$ ,  $v^{\text{EN}}=1$ ), and 146 discarded problems where the teacher failed in both languages (7,327 retained in total). The bridge rate ( $920/7,473 = 12.3\%$ ) closely tracks the 14.3% Korean teacher failure rate; the residual 2.0% gap is accounted for by these discards. We denote the resulting fine-tuned model **E**; all baselines (Section 4.1) share the student backbone, LoRA configuration, and training hyperparameters, differing only in dataset composition.

**Design choices.** Three design decisions distinguish DALR from related distillation recipes. (i) *Per-problem one-hot routing.* A bilingual mix (Setup D) pools Korean and English chains in a 50:50 ratio without regard to per-problem difficulty, so the choice of reasoning language is undirected. DALR instead makes a per-problem decision and emits exactly one target, the higher-quality language, for each question; it is strictly smaller than D (7,327 vs. 12,814), yet outperforms it (Section 4.5). (ii) *Teacher correctness as the difficulty signal.* Correctness is label-free, requires no extra annotation, and generalizes to any teacher–student pair. It also has a direct interpretation: if the strongest available model cannot produce a verifiable Korean solution, the Korean CoT for that problem is almost certainly low-quality reasoning, while the English CoT is verified. (iii) *Discard rather than fall back to a wrong chain.* For the 146 problems where both teacher languages fail, DALR removes the example outright instead of training on a known-wrong chain, a small (2%) cost in coverage in exchange for cleaner gradients.

### 3.3 Cross-Lingual Self-Consistency (XLSC)

DALR produces a model that reasons reliably in either Korean or English, but a single greedy decode commits to one chain in one language and inherits its errors. XLSC removes this commitment by sampling in both languages from the *same* DALR-trained model and aggregating across them, exploiting the fact that Korean and English chains traverse different intermediate steps for the same problem and produce partially decorrelated errors (Section 5 quantifies this).

**Aggregation rule.** For each Korean test problem  $q$ , we draw  $N$  chains under each system prompt  $s^L$ ,  $L \in \{\text{KO}, \text{EN}\}$ , from the same DALR-trained model  $\mathcal{M}_E$  at sampling temperature  $T$ , and return the plurality answer over the resulting  $2N$  extracted numbers:

$$c_k^L \sim \mathcal{M}_E(\cdot \mid s^L, q; T), \quad \hat{y}_{\text{XLSC}}(q) = \text{mode}(\{\text{extract}(c_k^L)\}_{L \in \{\text{KO}, \text{EN}\}, k=1, \dots, N}), \quad (2)$$

where  $\text{extract}(\cdot)$  recovers the final numeric answer from a chain and  $\text{mode}(\cdot)$  returns the most frequent element (plurality vote) over the  $2N$ -element multiset.

**Bilingual extraction.** Because samples come in two languages but vote jointly, the answer extractor must succeed on both: we apply two regular expressions in parallel, one matching the Korean answer-marker phrase and one matching its English equivalent (**answer is X**), with a fallback to the last number in the chain. Without this bilingual matching, chains ending in the other language’s answer line would be silently dropped from the vote.

**Design choices.** We use a single  $\mathcal{M}_E$  rather than two monolingual models—DALR already internalizes both languages—and weight all  $2N$  votes equally to avoid reintroducing the per-language reliability gap that DALR was designed to neutralize. XLSC costs  $2N$  forward passes per problem versus 1 for greedy decoding, purely at inference.

## 4 Experiments

### 4.1 Baselines

We design baselines to isolate one factor at a time. **A (Base)** is unmodified Qwen2.5-3B-Instruct, the no-training floor. **B (English-only)** and **C (Korean-only)** fine-tune on the 7,283 English or 6,407 Korean teacher chains respectively, the two “obvious” monolingual choices. **D (Bilingual 50/50)** balances B and C 1:1 by sub-sampling English down to the Korean count (12,814 examples, 6,407 of each), exposing the student to both languages in equal proportion; D answers whether undirected bilingual data alone suffices. **E\_random** is a tightly matched control for DALR: identical training pipeline, identical 920 English chains, but the English chains are injected on *randomly chosen Korean-solved* problems—i.e. on problems where the Korean teacher already succeeded, the opposite of DALR’s routing principle—while preserving the total of 7,327 examples. If DALR’s gain were data scaling, E and E\_random would tie; if it is routing, E should dominate. All conditions share the student backbone, LoRA configuration ( $r = 32$ ,  $\alpha = 64$ ), and training hyperparameters, differing only in dataset composition.

### 4.2 Data

**Training.** GSM8K [2] provides 7,473 training and 1,319 test problems in English grade-school mathematics. We use GSM8K-ko, a machine-translated Korean version of the training split obtained from the Hugging Face Hub. The teacher model (Gemini 3 Flash) generates chain-of-thought solutions independently in English and Korean; we retain only solutions whose final numeric answer matches the gold label (English: 7,283 valid out of 7,473; Korean: 6,407 valid). Invalid Korean CoTs are replaced by the corresponding English CoT in the DALR setup, or discarded otherwise.

**Evaluation.** We evaluate on **HRM8K** [3], a high-quality human-reviewed Korean translation of the GSM8K test set (1,319 problems), and on the original **GSM8K** English test

| Setup                          | HRM8K (KO)   | 95% CI (KO)    | GSM8K (EN) |
|--------------------------------|--------------|----------------|------------|
| A (Base zero-shot)             | 53.90        | [51.02, 56.48] | 71.65      |
| B (English CoT)                | 54.44        | [51.78, 57.32] | 74.22      |
| C (Korean CoT)                 | 59.51        | [56.63, 62.17] | 70.74      |
| D (Bilingual mix)              | 58.98        | [56.25, 61.79] | 74.98      |
| E (DALR)                       | 61.79        | [59.29, 64.44] | 69.37      |
| E_random (ablation)            | 58.83        | [56.10, 61.79] | 72.71      |
| E + KO×6 self-consistency      | 69.14        | [66.87, 71.49] | –          |
| E + XLSC ( $N=3+3$ , $T=0.7$ ) | <b>69.90</b> | [67.77, 72.19] | –          |

Table 1: Main results on HRM8K (Korean, 1,319 problems) and GSM8K (English, 1,319 problems) with the Qwen2.5-3B student. E (DALR) is the best single-model setup; **E + XLSC** achieves the best overall HRM8K accuracy (+**16.00** pts over the zero-shot base A, +**8.11** pts over E). XLSC accuracy is the mean over all  $\binom{6}{3} = 20$  subset compositions, each pairing 3 of the 6 Korean samples with the 3 English samples per problem. KO×6 is the matched-budget monolingual self-consistency ablation. 95% CIs are bootstrap intervals (1,000 resamples) over HRM8K problems; for XLSC, each problem’s correctness is first averaged across the 20 subsets before bootstrapping.

set (1,319 problems). HRM8K is our primary benchmark for two reasons. First, its translation quality is substantially higher than the machine-translated GSM8K-ko. Second, DALR fine-tunes the student exclusively on Korean questions; XLSC on English questions would require Korean reasoning about English problems, an out-of-distribution combination that pilot evaluation confirmed degrades accuracy. We therefore report XLSC results only on HRM8K.

### 4.3 Evaluation method

We report exact-match accuracy on the final numeric answer, extracted by a regular expression and compared to the gold answer with a tolerance of  $10^{-2}$ . Single-model evaluation uses greedy decoding (temperature 0, repetition penalty 1.1) with a zero-shot CoT system prompt in the target language. XLSC uses sampling with temperature  $T = 0.7$  and  $N = 3$  samples per language (6-way vote). We report 95% percentile bootstrap confidence intervals (1,000 resamples) on accuracy and paired McNemar tests for model comparisons.

### 4.4 Experimental details

The main student is **Qwen2.5-3B-Instruct**, fine-tuned with LoRA [14] ( $r = 32$ ,  $\alpha = 64$ , applied to all attention and MLP projection layers) for 3 epochs at learning rate  $2 \times 10^{-4}$ , effective batch size 8 (per-device 1, gradient accumulation 8), cosine schedule, with 4-bit quantization. We additionally train **Llama-3.2-3B-Instruct** and **EXAONE-3.5-2.4B-Instruct** with the same recipe (with module names adapted to each architecture) to assess robustness across student backbones.

### 4.5 Results

| Student model   | C     | D     | E (DALR) | E + XLSC     |
|-----------------|-------|-------|----------|--------------|
| Qwen2.5-3B      | 59.51 | 58.98 | 61.79    | <b>69.90</b> |
| Llama-3.2-3B    | 56.86 | 55.50 | 57.54    | <b>65.73</b> |
| EXAONE-3.5-2.4B | 60.35 | 60.20 | 60.50    | <b>69.29</b> |

Table 2: Robustness across student models on HRM8K (%). DALR is competitive with the Korean-only baseline C on all three students (largest gain on Qwen, +2.28 pts); XLSC on top of DALR adds a further +8.11 to +8.79 pts across architectures. For Qwen, the XLSC value is the mean over 20 subset compositions (Table 1); for Llama-3.2-3B and EXAONE-3.5-2.4B, we report single-draw measurements (one specific KO×3 + EN×3 seed).

**DALR improves Korean, and routing (not data) is the cause.** On HRM8K, E (61.79) exceeds the Korean-only baseline C (59.51) by 2.28 points and the bilingual mix D (58.98) by 2.81 points, becoming the best single model. The ablation E\_random uses the *same* 920 English bridges but injects them on Korean-solved problems instead of Korean-failed ones, and drops to 58.83 (*below* C), a 2.96-point gap from E (McNemar  $p = 0.030$ ). This isolates *routing*, rather than added data, as the source of DALR’s gain. Section 5 shows the gap is concentrated on exactly the problems where the quality differential should matter most.

**A Korean–English trade-off.** DALR raises Korean accuracy but slightly lowers English (E: 69.37 vs. C: 70.74). This is consistent with English bridges being concentrated on *hard* Korean problems, which exposes the model to English reasoning only in Korean question contexts and slightly weakens its performance on plain English problems. The trade-off motivates inference-time aggregation across both languages.

**XLSC further improves Korean accuracy.** Applying XLSC ( $N=3$  Korean +  $N=3$  English samples,  $T=0.7$ ) to the same E checkpoint lifts HRM8K from 61.79 to **69.90** (mean over 20 subset compositions; +**8.11** pts over E, +**16.00** pts over the zero-shot base A). To verify that this gain is not solely an artifact of voting over more samples, we ran matched-budget monolingual self-consistency: pure KO $\times$ 6 on the same E checkpoint reaches 69.14, accounting for most of the lift. Cross-lingual mixing still contributes an additional +**0.76** pts, and the direction is consistent—every one of the 20 (3+3) subset draws we tested exceeds pure KO $\times$ 6 in accuracy (sign test  $p < 10^{-5}$ ). Since E was trained predominantly on Korean data (87% KO, 12.6% EN bridges) and HRM8K problems are exclusively Korean, pure Korean sampling could plausibly have sufficed; the consistent positive lift from cross-lingual mixing instead indicates a real, if modest, decorrelation signal that single-language self-consistency does not capture.

**Robustness across students.** DALR is competitive with the Korean-only baseline C on all three students; the gap is largest on Qwen (+2.28 pts), modest on Llama (+0.68 pts), and effectively zero on EXAONE (+0.15 pts). XLSC on top of DALR adds a much larger and more consistent improvement across all architectures, ranging from +8.11 to +8.79 pts (Table 2), suggesting the inference-time benefit is not specific to Qwen.

## 5 Analysis

**Why does routing matter?** E and E\_random differ only in which 920 Korean training items are *replaced* by an English CoT; total volume, hyperparameters, and the English chains themselves are identical. The two replacements, however, are asymmetric in quality. E swaps an unreliable Korean chain (the teacher could not solve it) for a reliable English one (the teacher did), raising the average target quality at the swapped positions. E\_random does the opposite: it swaps a reliable Korean chain for an English chain on a problem the Korean teacher already solved, losing native-language supervision without gaining any new capability. The empirical signs mirror this: E\_random falls 0.68 points *below* the Korean-only baseline C (58.83 vs. 59.51), while E rises 2.28 points above it—a 2.96-point gap (McNemar  $p = 0.030$ ) attributable entirely to the placement of those 920 bridges.

**Where does DALR help?** We stratify HRM8K problems by the number of baselines (A, B, C, D) that solve them and report each method’s accuracy per bucket in Table 3. On *hard* problems that none of the four baselines solve (231 problems, 17.5%), C and D both score 0% by definition; E (DALR) recovers **16.88%** of them (39 problems), while E\_random reaches only 13.85% (32 problems), indicating that the English bridges do carry transferable reasoning patterns specifically when Korean reasoning fails. On *medium-difficulty* problems where some but not all baselines succeed (688 problems, 52.2%), E outperforms C by +2.62 points and E\_random by +3.78 points, confirming that routing rather than data volume drives the gain. On *easy* problems solved by all four baselines (400 problems, 30.3%), C is at 100% by construction and E drops to 93.25% (i.e. loses 27 of 400 problems)—the modest cost of reallocating training capacity to bridges. The qualitative picture matches DALR’s

design: bridges are placed where Korean reasoning is unreliable, and that is also where their leverage is largest.

| Group                        | #prob       | C      | D      | E (DALR)     | E_random |
|------------------------------|-------------|--------|--------|--------------|----------|
| Hard (0 baselines solve)     | 231 (17.5%) | 0.00   | 0.00   | <b>16.88</b> | 13.85    |
| Medium (1–3 baselines solve) | 688 (52.2%) | 55.96  | 54.94  | <b>58.58</b> | 54.80    |
| Easy (all 4 baselines solve) | 400 (30.3%) | 100.00 | 100.00 | 93.25        | 91.75    |

Table 3: HRM8K accuracy (%) by difficulty bucket, defined by the number of baselines (A, B, C, D) that solve each problem. DALR’s gain is concentrated on the hard subset, where Korean teacher reasoning fails.

**Why does XLSC work?** The Korean-only and English-only majorities at  $N=3$  are each 63.53% accurate on HRM8K—numerically identical, despite using disjoint samples in different languages. Combining them into a 6-way vote lifts mean accuracy to **69.90** (over 20 subset compositions; +6.37 pts over either single language at  $N=3$ ), evidence that the two languages produce partially decorrelated errors. The agreement structure at  $N=3$  makes this concrete: KO and EN majorities agree on 60.2% of problems, where the combined vote is correct 90.55% of the time; they disagree on 39.8%, and on this disagreement subset the KO and EN majorities are each right 22.7% of the time (neither is right on 54.7%). Decomposing the +6.37 lift into sample-budget and cross-lingual components, matched-budget monolingual self-consistency (KO $\times$ 6) reaches 69.14, accounting for +5.61 of the gain; the isolated cross-lingual contribution is **+0.76** pts. Small in magnitude but directionally unambiguous (20 of 20 subset draws favor XLSC over KO $\times$ 6, sign test  $p < 10^{-5}$ ), this residual lift confirms that cross-lingual aggregation captures error decorrelation that single-language self-consistency cannot.

**Statistical robustness.** All accuracies in Table 1 are reported with 95% percentile bootstrap intervals (1,000 resamples). Paired McNemar tests on HRM8K confirm the main claims: E vs. E\_random  $p = 0.030$ ; E vs. D  $p = 0.045$ ; E + XLSC vs. E  $p < 0.001$ . On GSM8K, E shows a significant drop relative to D ( $p < 0.001$ ) and E\_random ( $p = 0.016$ ), confirming the deliberate Korean–English trade-off described above.

## 6 Conclusion

We studied how to improve Korean mathematical reasoning in small language models by leveraging stronger English reasoning at two points in the pipeline. Our main contribution **DALR** routes teacher CoT language per problem according to teacher correctness; a controlled ablation shows the benefit comes from *routing* rather than added data, and the gain is concentrated on the hard problems where the routing decision matters most. **XLSC** exploits the bilingual capability instilled by DALR at inference time by sampling in both languages and majority-voting, contributing a further +8.11 to +8.79 pts across student architectures. A matched-budget ablation on Qwen shows that most of this gain reflects sample budget, with a small but consistent cross-lingual contribution (+0.76 pts; 20 of 20 subset draws favor XLSC over monolingual KO $\times$ 6 self-consistency).

**Limitations.** HRM8K is a high-quality translation of GSM8K rather than natively Korean math, so gains may partly reflect translated-problem solving; the DALR-vs-Korean-only margin is modest in absolute terms and attenuates on other student backbones; we use a single teacher (Gemini) and three small (2.4B–3B) students. Future work includes natively Korean math benchmarks, larger students, and adaptive sample budgets that allocate XLSC compute only to problems where languages disagree.

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## Appendix: Team contributions

- 이윤제 (2022320317): DALR design and implementation, E and E\_random training and evaluation on Qwen, XLSC implementation and evaluation, statistical analysis (bootstrap CIs, McNemar tests), paper writing (Sections 3.2 and 4.5).
- 김상준 (2022320306): Baseline (A–D) training and evaluation on Qwen, EXAONE-3.5-2.4B robustness experiments (full pipeline), paper writing (Section 3.3 and Analysis).
- 원준서 (2022320302): Teacher CoT generation and data construction, Llama-3.2-3B robustness experiments (full pipeline), paper writing (Experiments section).